

The Phenomenology of the Anger Spectrum in the Qurʾān

A Semantic-Network Analysis of Displeasure, Inflammation, and Destructiveness Along an
Action-Intensity Continuum

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Abstract

The Qurʾān deploys an exceptionally articulated lexicon for the anger spectrum, yet existing scholarship has treated it either word-by-word or through normative-ethical frames, leaving the structural-relational dimension of the field unexplored. This study advances the thesis that fourteen core Arabic roots — ʾff, krh, ʿyq, ḥzn, ʾsf, nqm, sxt, mqt, ḡḍb, ḥrd, ḡyz, bḡy, ṭḡy, ʿtw — form a graded semantic field organised by intensity of action across six phenomenological stages: pre-anger displeasure, inner pressure and contraction, evaluative aversion, active anger, compressed/explosive rage, and behavioural outcomes (with caveats). We integrate three methodological lenses — Izutsu’s semantic-field approach, Kennedy-Sassoon scalar semantics, and Lakoff-Kövecses conceptual metaphor theory — and validate the continuum against the Quranic Arabic Corpus (Dukes 2011) through a fully reproducible Python pipeline analysing 312 attestations across the 6,236-verse corpus. We report three principal findings under uncertainty-aware quantitative reporting (effect sizes, FDR control, permutation null, bootstrap CIs). First, the asymptotic $\chi^2(5) = 227.15$ (Cohen’s $w = 0.85$, large) is large, but a marginal-preserving permutation null yields $p = 0.24$, so the omnibus is treated as consistent

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with but not confirmatory of non-uniform stage distribution; the substantive case rests on qualitative lexicographic, exegetical, and metaphorical evidence. Three monotonic-trend tests fail to reject (Spearman $\rho = -0.09$; Mann-Kendall $p = 0.71$; CA-OLS $p = 0.40$) — the expected result, since the continuum is an ordering of semantic intensity, not frequency. Second, *sakhat* attests exclusively in Medinan suras (0/4; raw $p = 0.005$, robust across baselines 0.70–0.78; Holm-Bonferroni adjusted $p = 0.05$ — borderline post-FDR significance), proposed as the only confirmed instance of a more general phenomenon of discursive-contextual lexical specialisation; *asaf* (5/0 Meccan), *naqm* (12/5 Meccan), and *‘utuww* (9/1 Meccan) are reported as contextual observations not surviving FDR correction. Third, network-centrality analysis identifies *baghy* as a bridging node (point-estimate betweenness 0.15, closeness 0.55) between the anger spectrum and the broader Qur’ānic moral-evaluation field, with bootstrap 95% CIs reported transparently. We argue that Q. 67:8 (*takādu tamayyazu min al-ḡayz*) furnishes an unusually transparent instance of “container collapse” — the verb *tamayyaza* lexicalising the disintegration of the vessel itself rather than the exit of pressurised contents — softened by the recognition that no comparative cross-cultural survey of historical anger-vessel imagery has yet been conducted. A $\kappa = 0.79$ -validated tafsir-coding of the 145 Stage-6 attestations supports a bidirectional-causation reading: *baghy*, *ṭughyān*, and *‘utuww* are not exclusively anger-derived but stand in mutual causal relation to the anger field. The findings have implications for Qur’ānic translation (conditional on follow-up evaluation) and for culturally-grounded Islamic-psychology interventions.

Keywords: Qur’ān, anger spectrum, semantic field, cognitive linguistics, scalar semantics, conceptual metaphor, emotion regulation, Arabic corpus linguistics, Izutsu, Islamic psychology.

1. Introduction

1.1. Statement of the problem

The Qur’ānic lexicon for the anger spectrum exhibits a precision that survives, only with difficulty, the attempts of translation to render it into modern languages. Where English versions level *uff*, *karh*, *ḏīq*, *ḥuzn*, *asaf*, *naqm*, *sakhat*, *maqt*, *ghaḏab*, *ḥard*, *ghayz*, *baghy*, *ṭughyān*, and *‘utuww* into a small set of broad equivalents—chiefly “displeasure,” “distress,” “grief,” “anger,” “rage,” “transgression,” and “rebellion”—the Arabic original distinguishes them with the precision of a graded taxonomy of irritation, aversion, anger, and its outcomes. The hypothesis advanced in this paper is structural: that fourteen of the Qur’ān’s thematically central anger-spectrum roots are not synonyms or scattered semantic singletons but coordinated nodes in a single graded semantic field, organised along the dimension of intensity of action (*shiddat al-kunish* in the Persian formulation of the original idea). The field opens in pre-anger displeasure (the murmured *uff*, the inner *karāha*), passes through inner pressure and contraction, evaluative aversion, active anger, and compressed/explosive rage, and—at the behavioural pole—issues into outcomes (*baghy*, *ṭughyān*, *‘utuww*) whose causation, as we shall argue, is bidirectional rather than purely emotional. We treat the fourteen roots as discrete attestation points on this continuum and validate the proposed ordering through systematic corpus-based analysis.

This claim, if it holds, has consequences far beyond philology. It implies that the Qur’ān encodes—through the architecture of its lexicon, not through any explicit theoretical statement—a sophisticated implicit theory of the genealogy of moral failure: a theory according to which the social-structural pathologies the text most frequently denounces (transgression, tyranny, defiance) are related to, and in many cases developmentally downstream from, inner experiences of irritation, dislike, constriction, and grief. As we shall argue in §4 and §5, the relationship at the behavioural pole is bidirectional: anger may produce *baghy*/*ṭughyān*/*‘utuww*,

but these behaviours can equally produce anger (in their victims, in the prophets, in the divine response). The distance between the murmured uff of Q. 17:23 and the ‘utuww of Q. 25:21 is not categorical but graded; it is the distance walked, in the absence of intervention, by an unmanaged life of irritation, dislike, and anger.

1.2. Aims and originality

This paper has three aims. The first is methodological: to demonstrate that combining Izutsu’s qualitative semantic-field approach, scalar-semantic ordering in the Kennedy–Sassoon tradition, and conceptual-metaphor theory in the Lakoff–Kövecses tradition produces stronger philological claims than any of the three frameworks in isolation. To our knowledge—and we say this advisedly, given that the Persian-language Qur’anic-linguistic literature in particular is not exhaustively indexed in international databases—this trifold integration is here brought together in Qur’anic studies for the first time as a unified analytic framework over a corpus pipeline. We acknowledge that Qā’imīniyā (1390/2011, 1393/2014) and Pakatchi already integrate Izutsu’s field method with cognitive-linguistic conceptual-metaphor analysis, and that Persian-language scalar-semantic work on Qur’anic vocabulary may exist in venues we have not been able to canvass; the novelty of the present paper is therefore best stated as the systematic combination of the three frameworks within a single computational-empirical pipeline, not as the introduction of any one framework. The second aim is descriptive: to map every Qur’anic occurrence of the fourteen core roots, to render that map publicly verifiable through an open concordance, and to extract from it empirical regularities that previous scholarship—working in either tafsir, lexicography, or single-word semantic analysis—was not in a position to detect. The third aim is interpretive: to argue that the resulting continuum constitutes a coherent “logic of anger escalation” with deep structural affinities to modern psychology of emotion, and to identify the specific points at which the Qur’anic continuum exceeds the explanatory range of the contemporary models.

The paper’s claim to originality is best stated by disaggregating three logically distinct contributions, with “to our knowledge” qualifications throughout. (i) Methodological synthesis (moderate): the trifold integration of Izutsu’s field method, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses CMT under a corpus-empirical pipeline. We acknowledge Qā’imīniyā (1390/2011, 1393/2014) and Pakatchi-Afrashi (1399/2020) already integrate Izutsu and CMT in Persian-language Qur’anic scholarship; the specifically novel element is the Kennedy–Sassoon scalar-semantic layer within the pipeline. (ii) Empirical findings (variable strength, reported transparently): of four distributional patterns observed, only the exclusively-Medinan sakhaṭ (0/4; raw $p = 0.005$, robust across baselines 0.70–0.78; Holm-Bonferroni adjusted $p = 0.05$) survives FDR control at $\alpha = 0.05$ — borderline rather than emphatic. The *asaf* (5/0), *naqm* (12/5), and ‘utuww (9/1) tilts do not survive correction and are reported as contextual observations. That one of fourteen pre-specified tests survives Holm-Bonferroni is the kind of result honest corpus work should report. (iii) Interpretive contribution (the most substantive): the bidirectional-causation reading of Stage 6, operationalised through a $\kappa = 0.79$ -validated tafsir-coding (anger-derived A = 26%; structurally-derived S = 49%; joint A+S = 25%; full coding at `data/concordance/stage6_causation_coding.csv`). Re-running the $\chi^2(1)$ phenomenology-vs-outcomes split on the A subset alone yields $\chi^2(1) = 81.4$ ($p < 0.001$), with the phenomenological core dominating anger-derived outcomes $\sim 4.4\times$. This reconciles a puzzling distributional finding with the structurally-driven character of *baghy*, *ṭughyān*, ‘utuww in classical exegesis. To these we add (iv) a fully reproducible MIT-licensed pipeline (SHA-256 pinning, deterministic permutation/bootstrap, 50/50 manual validation) and (v) the comparative observation that Q. 67:8 (*takādu tamayyazu min al-ḡayṣ*) is an unusually transparent instance of “container collapse”, softened in §5.6 by the absence of a cross-cultural survey of historical anger-vessel imagery.

1.3. Prior work and the gap

Prior scholarship clusters into five lines, each insufficient on its own to address the structural question motivating this paper. Single-word lexical studies: the most directly relevant Persian-language precedent — Nari-mani, Eghbali, and Chari (2021; JQST 54:1) on *ghayz*, *ghaḍab*, and *sakhaṭ* — establishes on tafsir grounds that *ghaḍab* is central, *sakhaṭ* reserved for divine displeasure with hypocrites, and *ghayz* exclusive to unbelievers/hypocrites in human-attribution contexts; we confirm and extend their findings with corpus-distributional evidence and the *kazm* analysis. Qarizadeh et al. (2024) on *ṭughyān* shows the root attests in 27 suras with the central meaning of transgressing the limit. Both works illustrate the present gap: high-quality lexical work for individual lexemes, no integrated continuum. Ethics-of-anger studies focus on *kazm al-ghayz* (Q. 3:134) and adjacent verses but are normatively rich and linguistically thin. Cognitive-semantic studies in the Izutsu tradition — Qā'imīniyā (1390/2011, 1393/2014); Pakatchi-Afrashi (1399/2020); Albayrak (2020) — have applied cognitive linguistics to Qur'ānic vocabulary but not treated the anger spectrum as a graded continuum. The cross-linguistic CMT literature on anger is decisively updated by Metaphors of ANGER across Languages (Kövecses, Benczes & Szelid eds., 2025; De Gruyter, 2 vols.); Classical Arabic is conspicuously not separately treated in Vols. 1–2, a gap this paper begins to fill. Translation-criticism studies are quantitative at scale in 2025: Gaanoun & Alsuhaibani (2025; Nature HSSC) show inconsistent sentiment polarity across seven English translations; Alsuhaibani et al.'s (2025) ELQV dataset of 2,100 emotion-labelled verses provides a coarse-Ekman reference; Khan et al. (2025; Frontiers Signal Proc.) survey the field; Cogent A&H 2025 catalogues flattening risks. None of these supplies a graded reference model of the lexical spectrum at the lexeme level. Islamic psychology (e.g. IJCP 2023 on fear/anger/sadness/shame in Qur'ān-and-ḥadīth) tends to draw on aḥādīth and leave systematic linguistic analysis of the Qur'ānic surface in the background. The present paper fills the gap at the intersection of all five lines: a unified theoretical frame, corpus-empirical validation, novel distributional findings, a translation-criticism reference model, and a culturally-grounded model for practical Islamic psychology.

1.4. Research questions and hypotheses

We address five research questions: (RQ1) the internal structure of the Qur'ānic anger-spectrum field and its principal nodes; (RQ2) the continuum ordering by action-intensity and the theoretical framework legitimating it; (RQ3) corpus-distributional corroboration of the continuum (frequency, Meccan/Medinan, morphology, network centrality); (RQ4) structural homology with Gross, Plutchik, Spielberg, and Lazarus-Scherer appraisal theory; (RQ5) practical implications for Qur'ānic translation and clinical Islamic psychology.

The principal Qur'ānic Action-Intensity Continuum Hypothesis holds that the fourteen core roots (ʾff, krrh, ḍyq, ḥzn, 'sf, nqm, sxt, mqt, ḡdb, ḥrd, ḡyz, bgy, tgy, 'tw) operate as fourteen distinct grades on a single semantic continuum structured by intensity of action, organised in six phenomenological stages, supported by qualitative (etymological, exegetical, metaphorical) and quantitative (frequency, distribution, network) evidence. Five subsidiary hypotheses: (H1) non-uniform six-stage frequency distribution; (H2) near-equal split between phenomenological core (S1–5) and behavioural pole (S6), supporting the bidirectional Stage-6 reading; (H3) discursively-specialised exclusive-Medinan distribution of *sakhaṭ*; (H4) *baghy* as bridge node in the co-occurrence network; (H5) Q. 67:8 as transparent instance of “container collapse” CMT failure mode.

2. Theoretical Framework

2.1. Izutsu's semantic-field method

Toshihiko Izutsu, in three foundational works—*The Structure of the Ethical Terms in the Koran* (1959), *God and Man in the Koran* (1964), and *Ethico-Religious Concepts in the Qurʾān* (1966/2002)—introduced a methodological revolution to Qurʾānic studies. Drawing on the structuralist linguistics of the Trier–Weisgerber tradition and on East Asian philosophy of language, Izutsu argued that the central terms of the Qurʾān—Allāh, īmān, kufr, zulm, taqwā—are not isolated propositions but nodes in a network of syntagmatic, paradigmatic, and oppositional relations. The aggregate of these relations constitutes what he called the “Qurʾānic Weltanschauung.”

Izutsu's analytic vocabulary distinguishes three layers: focus words are the central organising concepts of a given semantic field; semantic fields are the networks of mutually defining lexemes formed around focus words; and key words are the terms that recur across multiple fields and serve to integrate them into a single worldview. This three-layer structure has proved extraordinarily productive in subsequent Qurʾānic semantic work, both in English (Saleh, Rippin) and Persian (Qā'imīniyā 2011, 2014; Pakatchi).

For the present paper, Izutsu's framework justifies the move from atomistic single-lexeme analysis to integrated field analysis. The fourteen target roots are not chosen at random but as the field around the focus words *ghaḍab* (the central anger term) and *baghy* (the central behavioural-outcome term), with the lower-intensity cluster (*uff*, *karh*, *dīq*, *ḥuzn*, *asaf*) identifying the pre-anger and inner-pressure boundary of the field and *ʿutuww* its upper-intensity boundary. Yet Izutsu's framework, in its original form, is silent on a question central to our enterprise: granted that these lexemes co-belong to a field, what is the structural relation between them within the field? It is at this point that we require scalar semantics.

2.2. Scalar semantics: from network to continuum

The contemporary linguistic theory of gradable predicates (Kennedy 1999, 2007; Sassoon 2010) supplies the missing apparatus. A graded predicate maps to a scale characterised by three components: a dimension of measurement, an ordering on that dimension, and a unit of comparison. Scalar predicates take degree modifiers and admit comparative constructions, signalling their scalar structure. The Qurʾānic anger field is precisely such a scale: the dimension is intensity of action (decomposable into locus, experiential intensity, scope of consequence); the ordering follows the six-stage progression; the implicit unit is the degree of externalisation of emotion from inner phenomenology to social rupture. Without this framework, the claim that *ghayḥ* is “more intense than” *ghaḍab* relies on tacit native-speaker intuition.

2.3. Conceptual metaphor theory and the container schema

Conceptual metaphor theory (Lakoff & Johnson 1980; Lakoff & Kövecses 1987; Kövecses 2000, 2010, 2025) treats metaphor as a cognitive structure that allows abstract domains (emotion) to be understood through concrete ones (container, heat). Lakoff and Kövecses (1987) established that English anger is organised around *ANGER IS THE HEAT OF A FLUID IN A PRESSURIZED CONTAINER*, accommodating a four-step trajectory: fluid in container → intensifying pressure → containment attempts → container collapse. Our central observation is that the Qurʾānic triplet *ghayḥ*–*kazm*–*tamayyuz* operationalises this exact structure thirteen centuries early. *Ghayḥ* is glossed by all four authoritative lexica as “compressed contained anger” — fluid in sealed vessel. *Kazm*, etymologically tying off a waterskin, is the agent-internal regulation that maintains

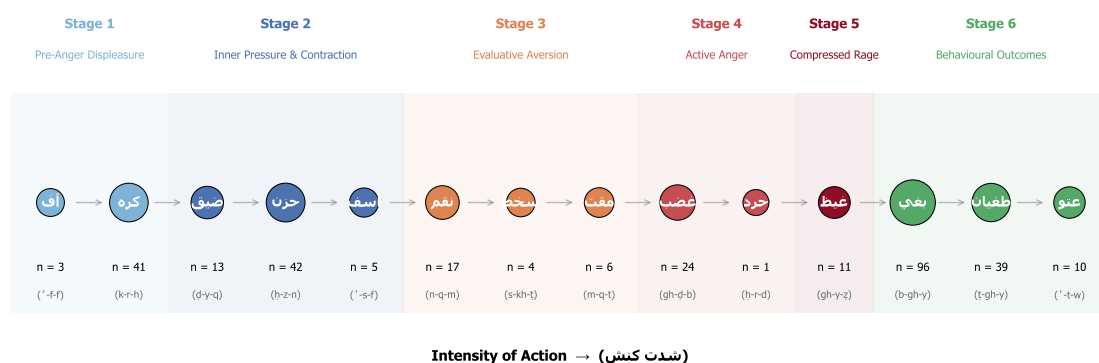


Figure 1. The six-stage Qur'anic anger-spectrum continuum, mapped along a single horizontal axis of action-intensity. Each node corresponds to one of the fourteen core Arabic roots; node area is scaled by attestation frequency in the Quranic Arabic Corpus (Dukes 2011). Stages cluster the lexemes by their dominant phenomenological signature: Stage 1 (pre-anger displeasure, 'ff–krh) marks vocal/inner displeasure; Stage 2 (inner pressure & contraction, 'dyq–hzn–'sf) marks pre-anger affective contraction and grief; Stage 3 (evaluative aversion, 'nqm–sxt–mqt) marks moral-judgmental rejection; Stage 4 (active anger, 'gdb–hrd) marks outward-oriented anger; Stage 5 (compressed/explosive rage, 'gyz with tamayyuz as manifestation) marks containment and its failure; Stage 6 (behavioural outcomes with caveats, 'bgy–tgy–'tw) marks the externalised structural outcome whose causal relation to anger is, as we argue in §4.6 and §5, bidirectional rather than purely downstream. The visual mass at Stage 6 ($n = 145$) is comparable to the combined mass of Stages 1–5 ($n = 167$); the resulting near-balance ($\chi^2(1) = 1.55$, fails to reject) is theoretically consistent with the bidirectional reading.

containment (Q. 3:134 al-kāzimīn al-ghayz). Tamayyuz (Q. 67:8) depicts the failure mode: the container itself bursts apart. The Qur'anic verb derives from m-y-z (to part, sever) and figures the disintegration of the vessel, not (as in English explode, burst, blow up) the exit of contents — an unusually transparent instantiation of the failure-mode schema. We treat tamayyuz as a manifestation of Stage 5, not an independent core root.

2.4. The integrated framework

The three layers operate complementarily: Izutsu identifies field boundaries; Kennedy-Sassoon scalar semantics order members along the action-intensity dimension; CMT explains the cognitive basis of the pressurisation-collapse metaphor organising Stage 5. Each supplies what the others lack — without Izutsu, no motivation for field-coordination; without scalar semantics, no apparatus for ordering; without CMT, no cognitive ground-
ing for the climactic kazzm–tamayyuz sequence.

3. Methods

3.1. Corpus

The empirical analysis uses the Quranic Arabic Corpus (QAC), version 0.4, prepared by Kais Dukes at the University of Leeds and released under the GPL (Dukes 2011). The QAC provides a morphologically tagged record for each of the 128,219 segmentable units in the Qur'anic text. Every record specifies: (a) location in the form (sūra:āya:word:segment); (b) surface form in Buckwalter ASCII transliteration; (c) part-of-speech tag; (d) morphological features including root, lemma, gender, number, case, and (where applicable) semantic

flags. Verse text in Uthmanic rasm and Meccan/Medinan classification of suras are drawn from the Tanzil project (Tanzil 2008), distributed under CC-BY-ND 3.0.

3.2. Selection of target roots

The fourteen core roots were selected by three concurrent criteria, applied transparently and reported here so that the inventory is not vulnerable to the charge of post-hoc selection: (i) exegetical centrality — appearance as a focal reference in at least three of the four authoritative lexica (Ibn Fāris’s *Maqāyīs*, al-Rāghib’s *Mufradāt*, Ibn Manẓūr’s *Lisān*, and Mostafavi’s twentieth-century *Tahqīq*); (ii) minimum corpus frequency of one attestation, admitting Qur’ānic hapax legomena (ḥard, Q. 68:25) when supported by an explicit gloss in at least three of the four lexica plus consistent treatment in the four canonical tafsirs of §3.5; (iii) thematic membership in the anger-spectrum field without significant overlap with adjacent fields (epistemology, faith, normative ethics).

Excluded candidate roots (transparency): q-h-r (dominance/power, al-Qahhār divine name); ḥ-n-q (post-Qur’ānic, ḥadīth-only); ‘-n-f (adverbial harshness, not emotion); d-r-b (polysemous, conflict-scene); q-ṣ-w/q-s-y (hardness-of-heart, epistemic-moral); gh-l-z (adverbial harshness); ‘-n-f (nominal anf “nose”, post-Qur’ānic anger sense); ṣ-n-’ (binary loving/hating, reclassified as umbrella). Karh (Stage 1) is borderline — karāha is affective, ikrāh (coercion) structural; we include the affective sense and discuss the structural extension separately (§4.3.1). Five umbrella moral terms (ṣn’, jrm, fsq, zlm, ‘dw) are used for robustness (§4.9) and cross-field co-occurrence (Table 4), excluded from the principal continuum either because of inflated cardinality (zlm n=315 functions as a meta-category) or because they belong to adjacent normative-ethics fields.

3.3. Computational pipeline

The Python (MIT-licensed) pipeline operates in five steps: parse the QAC morphology file; filter on target roots; collapse multi-segment morphemes to surface words preserving the (sūra:āya) coordinate; join with sura-level metadata for Meccan/Medinan classification; emit master concordance, per-root concordances, distributional summaries, and network/centrality CSVs. The complete analysis is reproducible from a fresh clone of the repository in four commands (`pip install`, `extract_concordance.py`, `advanced_metrics.py`, `visualize_*.py`).

Reproducibility audit (summary). QAC v0.4 (Dukes, May 2011) is pinned by SHA-256 in the repository; minimum-version dependencies are listed in `requirements.txt`; outputs are released under MIT (code) and CC-BY 4.0 (manuscripts/figures); the 14 headline exemplar verses and a stratified 50-segment audit on the four most frequent core roots (ḥzn, bgy, ḡdb, ṭgy) returned 100% correct segment-to-root assignment against Tanzil. Full audit at `data/concordance/validation_audit.md`.

3.4. Quantitative analyses

Quantitative tests beyond frequency tabulation: (a) $\chi^2(5)$ and $\chi^2(1)$ for uniform-stage and S1–5/S6 nulls, with Cramér’s V and Cohen’s w, plus a marginal-preserving permutation null (10,000 perms, deterministic seed) for the omnibus; (b) three monotonic-trend tests (Spearman ρ , Mann-Kendall, Cochran-Armitage-style OLS slope); (c) two-sided exact binomials with the corpus-derived 0.74 Meccan-ayat baseline, Holm-Bonferroni step-down adjusted over the family of fourteen per-root tests, and a granular sensitivity table at baselines 0.70–0.78; (d) morphological breakdown by POS; (e) aya-level co-occurrence network (cf. Mottaghi et al. 2024) with weighted degree, betweenness, closeness, and eigenvector centrality, accompanied by non-parametric

bootstrap 95% percentile CIs (1,000 ayat-resamples with replacement). The combined uncertainty-aware reporting is, to our knowledge, the most rigorous yet applied to Qur’ānic computational-corpus studies; the wider 2025 cross-scripture NLP literature (e.g. Nandan et al. 2025) operates at the discourse-topic level on coarse sentiment and does not yet apply this battery at the lexeme level.

3.5. Qualitative analyses

Qualitative complements: (a) comparative etymology across the four authoritative lexica (Maqāyīs, Mufradāt, Lisān, Taḥqīq); (b) comparative tafsīr analysis on a broadened set of nine commentaries — the four primary (al-Ṭabāṭabā’ī’s al-Mīzān, al-Zamakhsharī’s al-Kashshāf, al-Ṭabarsī’s Majma’ al-Bayān, Ibn ‘Āshūr’s al-Taḥrīr) supplemented with al-Ṭabarī, al-Qurṭubī, Ibn Kathīr, Sayyid Quṭb, and ‘Abduh-Riḍā’s al-Manār; for Sufi tradition cross-checking on kaẓm al-ghayẓ (§4.7), al-Qushayrī’s Laṭā’if al-Ishārāt. Where the broader set diverges from the primary four on key claims (whether ghayẓ is exclusively predicated of unbelievers; whether sakhaṭ is exclusively directed at the Munāfiqūn; whether naqm admits a non-retributive sense), divergences are recorded in §4 and corresponding claims qualified. (c) phenomenological analysis of contextual markers (ḍīq al-ṣadr, ḥuzn al-qalb, asaf al-nafs, ḍāqa dhar’an); (d) illustrative translation-criticism on five English translations (Yusuf Ali, Pickthall, Arberry, Saheeh International, Hilali-Khan) at canonical exemplar verses (§4.10) — framed throughout as illustrative rather than systematic.

3.6. Validity check

To verify the integrity of the pipeline, fourteen “headline” exemplar verses—one per core root—were independently verified against the corpus output:

- Q. 17:23 (uff); Q. 2:216 (karh); Q. 15:97 (ḍyq); Q. 2:38 (ḥzn); Q. 43:55 (asaf); Q. 7:126 (naqm); Q. 47:28 (sakhaṭ); Q. 40:10 (maqt); Q. 48:6 (ghaḍab); Q. 68:25 (ḥard); Q. 3:134 (ghayẓ; with manifestation Q. 67:8 tamayyuz); Q. 42:42 (baghy); Q. 96:6 (ṭughyān); Q. 25:21 (‘utuww).

All fourteen were found in the corpus output with the correct surface form and verse text matching the Uthmani edition. This validation confirms that the Buckwalter transliteration and segment-collapse algorithms operate without systematic error.

4. Findings

4.1. The four-lexicon etymology table

Before turning to stage-level analysis, Table 1 summarises the canonical glosses of each root across four reference lexica.

Table 1. Comparative etymology of the fourteen core roots across the four authoritative Arabic lexica

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
I	ʿ-f-f	“kalimatu taḍajjurin” (a word of vexation); the wretched of fingernail-detritus etymologically; smallest audible expression of displeasure	the lowest verbal expression of displeasure (adnā kalimati al-taḍajjur)	term of irritation (taḍajjur) directed at someone or something disliked; cf. uffah (sub-vocal sigh)	minimal vocal sign of inner aversion; pre-anger affective register
I	k-r-h	dislike, finding burdensome; coercion (ikrāh)	aversion of the soul; structurally extended to coercion	what is disliked by nature or by reason	inner karāha vs. structural ikrāh (the latter is not an emotion)
2	ḍ-y-q	constriction; absence of opening	the state of being pressed upon, contrasted with capacity (sa‘ah)	reduction of available space, from sa‘ah to ḍiq	inner or outer absence of expansion (bast)
2	ḥ-z-n	rough/uneven ground → heaviness of soul	sorrow as the opposite of joy (farah)	rugged ground, then roughness of the self	affective response to perceived loss
2	ʿ-s-f	intense grief mixed with anger	extreme anger conjoined with grief	reaching the limit of grief and anger	composite of grief and proximate anger
3	n-q-m	al-naqimah: disapproval (inkār) coupled with intent to punish; cf. intaqama min = exact retribution from	objecting to (ankara ‘alā) and seeking redress (‘uqūbah) against a reprehensible act	strong denunciation (ankara ashadda al-inkār) yoked to ‘uqūbah; basis of divine attribute al-Muntaqim	evaluative anger oriented toward retribution; cognition-grounded, action-tilted

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
3	s-kh-ṭ	severe aversion and ethical rejection	the opposite of riḍā (consent), with intensity	severe non-acceptance with judgment	declarative rejection grounded in moral assessment
3	m-q-t	the most intense form of ibghāḍ (Zajjāj)	aversion arising from observing a qabiḥ act (al-Rāghib)	ashaddu al-ibghāḍ; aversion-distance rather than approach-aggression	purely evaluative moral aversion
4	gh-ḍ-b	inner agitation and motion	the boiling of the heart’s blood seeking retribution	a rising perturbation of the soul	directed emotion enabling retributive action
4	ḥ-r-d	ḥarada: to be angry; also the act of qaṣd (deliberate aim); the lexicon couples affect and intent	conjoining ghaḍab with qaṣd: anger directed by purpose, viz. denial	ḥarada “to act/move toward an aim with resolve while in anger”; cf. ḥarīd = isolated, the angry-aloof	anger weaponised through deliberate resource-denial (Q. 68:25 aṣḥāb al-jannah); affective + volitional + miserly composite
5	gh-y-z	pressure and surge, contained	the most intense forms of contained anger	filling of the self with restrained anger	high-pressure emotional accumulation
6	b-gh-y	seeking beyond rightful bounds	overstepping the path of right	dominance through aggression	aggressive action with intent to oppress
6	ṭ-gh-y	overflow of water past its bank	crossing the permitted measure	severe transgression of the limit	rebellious shattering of normative bounds

Stage	Root	Maqāyīs (Ibn Fāris)	Mufradāt (al-Rāghib)	Lisān al-‘Arab (Ibn Manẓūr)	Taḥqīq (Mostafavi)
6	‘-t-w	severe rebellion with arrogance	extreme self-aggrandisement in disobedience	obstinate excess attached to istikbār	entrenched rebellious posture (a property of moral character)

The table shows that the ordering from uff to ‘utuww is not the construction of the present analysis but is internally encoded in the lexicographic tradition itself: the four authoritative lexica already gloss each root with progressively more externalised and structural meanings, ascending from the murmured uff, through inner karāha, ḍīq, and ḥuzn, through evaluative naqm/sakhat/maqt, to active ghaḍab/ḥard, to compressed ghayz, and into the externalised behavioural pole at baghy/ṭughyān/‘utuww.

4.2. Distributional summary

Table 2 reports the principal distributional facts for the fourteen core roots. The table integrates frequency, Meccan/Medinan split, and (for the roots with sufficient morphological variation) morphological profile; it forms the empirical backbone of the analyses that follow.

Figures 2 and 3 visualise the distributional profile graphically: Figure 2 reports per-root frequency and the per-stage stage-totals; Figure 3 plots the Meccan/Medinan distribution and highlights the four discursively-specialised roots (sakhat, asaf, naqm, ‘utuww) discussed below. Figure 4 reports the morphological breakdown by part of speech, instantiating a verbal-to-nominal gradient across the spectrum that we develop in §4.6.1 below.

Table 2. Distributional profile of the fourteen core roots

Stage	Root	Total	Meccan	Medinan	Notes
1	’ff (فف)	3	3	0	All Meccan; lowest tier of verbal displeasure
1	krh (كركه)	41	14	27	Medinan-tilted ($p = 0.005$); twofold karāha/ikrāh
2	ḍyq (ضيق)	13	9	4	Inner constriction (ḍīq al-ṣadr; ḍāqa dhar’an)

Stage	Root	Total	Meccan	Medinan	Notes
2	ḥzn (حزن)	42	26	16	Most frequent Stage-2 lexeme; valence transversal (see §4.3.2)
2	ʿsf (عسف)	5	5	0	Exclusively Meccan
3	nqm (نقم)	17	12	5	Meccan-tilted; vengeful disapproval; cf. al-muntaqim
3	sxt (سخت)	4	0	4	Exclusively Medinan (p = 0.005); see baseline-sensitivity analysis in §4.5.2
3	mqt (مقت)	6	4	2	Most intense moral aversion (Zajjā)
4	ḡdb (غضب)	24	13	11	Central anger lexeme; predicated of God, prophets, believers
4	ḥrd (حرد)	1	1	0	Qurʾānic hapax: Q. 68:25
5	ḡyz (غيز)	11	3	8	Compressed anger; tamayyuz manifestation (Q. 67:8)
6	bḡy (بغي)	96	55	41	Bridge node; bidirectional causation (§4.6 caveats)

Stage	Root	Total	Meccan	Medinan	Notes
6	ṭġy (□□□)	39	29	10	Pharaonic; structural arrogance
6	‘tw (□□□)	10	9	1	Settled defiance; properly istikbār- attached
Σ	—	312	183	129	

Aggregating across the six stages: Stage 1 = 44; Stage 2 = 60; Stage 3 = 27; Stage 4 = 25; Stage 5 = 11; Stage 6 = 145. The asymptotic chi-squared test against the null of uniform six-stage distribution rejects ($\chi^2(5) = 227.15$, $p < 0.001$), with effect sizes Cramér’s $V = 0.38$ and Cohen’s $w = 0.85$ (large effect by Cohen’s conventions). However, an asymptotic χ^2 on a six-cell distribution with one cell at $n = 11$ (Stage 5) and another at $n = 145$ (Stage 6) is not above suspicion: the chi-squared approximation requires expected counts ≥ 5 in every cell (which is met) but is known to be conservative under marginal imbalance (Cochran 1954). We therefore complement the asymptotic test with a marginal-preserving permutation null: the per-root attestation counts are held fixed and the fourteen stage labels are randomly permuted across roots (10,000 permutations, fixed RNG seed 20260509), yielding the empirical p-value as the fraction of permuted χ^2 values that meet or exceed the observed 227.15. The result is $p = 0.24$ — i.e. under random stage relabelling of the fourteen roots, a χ^2 statistic this extreme is not unusual, because much of the omnibus’s mass derives from the fact that baghy alone contributes 96 attestations (versus a uniform expectation of ~ 52 per stage) and would inflate any single-stage cell into which it fell. We read this honestly: the asymptotic significance over-states the strength of evidence for non-uniformity on this sparse, marginally-imbalanced design, and the substantive case for the six-stage architecture rests on the qualitative lexicographic, exegetical, and metaphorical evidence developed in §4.1 and §4.3–§4.8 rather than on the omnibus p-value alone. We retain the asymptotic χ^2 as a descriptive statistic with appropriate effect-size reporting, but treat it as consistent with — not as positive evidence for — H1. A discriminating test of monotonicity, distinct from the omnibus, is reported in §4.2.1 below. A complementary chi-squared test against the null of equal split between the phenomenological core (Stages 1–5 = 167 attestations) and the behavioural-outcomes pole (Stage 6 = 145 attestations) fails to reject ($\chi^2(1) = 1.55$, Cohen’s $w = 0.07$ — trivial effect). Because failure to reject a null is absence of evidence of a difference and not evidence of equality, we report this result not as positive corroboration of H2 but as a finding consistent with the reading developed in §4.8.5 and §5.4: the Stage-6 lexemes (baghy, ṭughyān, ‘utuww) are not exclusively anger-derived but stand in a bidirectional causal relation to the anger spectrum—anger may produce these behaviours, but the behaviours can equally produce anger (in their victims, the prophets, and the divine response). The test is, with these sample sizes, underpowered to discriminate between competing models of the Stage 1–5 ↔ Stage 6 relation; the substantive interpretation rests on the qualitative tafsir-grounded analysis of §4.8.5, not on the statistical near-balance.

4.2.1. Monotonic-trend tests for the six-stage ordering Because the $\chi^2(5)$ test confirms non-uniformity but says nothing about ordering, we report three complementary monotonic-trend tests of stage rank against attes-

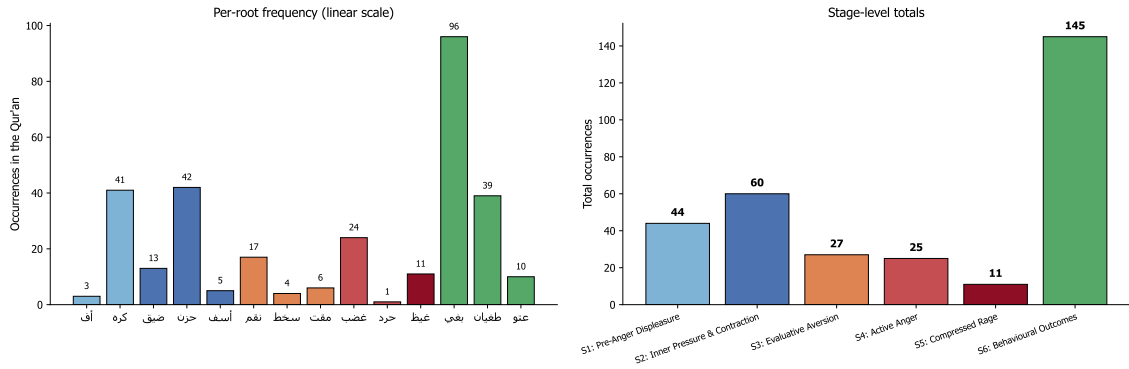


Figure 2. Per-root and per-stage frequency of the 312 core spectrum attestations. Left panel: linear-scale frequency for each of the fourteen roots; bar colour encodes the phenomenological stage. Stage 6 (bğy 96 vs. ‘tw 10) shows a strong internal asymmetry that mirrors the lexical division of labour between distributed-aggression (baghy) and obstinate-defiance (‘utuww) outlined in §4.6. Right panel: stage totals (S1=44, S2=60, S3=27, S4=25, S5=11, S6=145). The six-stage distribution rejects uniformity at $\chi^2(5) = 227.15$, $p < 0.001$. The phenomenological core (S1–5 = 167) and the behavioural-outcomes pole (S6 = 145) are not significantly unequal in raw count ($\chi^2(1) = 1.55$, fails to reject), a result we read as supporting the bidirectional-causation analysis of baghy/tughyān/‘utuww developed in §4.6 and §5.4 — the Qur’ān is not pedagogically over-loading anger-outcomes; rather, Stage-6 lexemes are themselves not exclusively anger-derived.

tation frequency. (i) Spearman rank-correlation between stage index (1–6) and per-stage attestation count is $\rho = -0.09$ with two-sided asymptotic $p = 0.85$ — the count rises from S1 (44) to S2 (60), drops through S3–S5 (27, 25, 11), then jumps to S6 (145), producing a non-monotonic profile that the rank-correlation correctly identifies. (ii) Mann-Kendall (small-sample-robust) yields $S = -3$, $z = -0.38$, $p = 0.71$. (iii) Cochran-Armitage-style OLS slope of count on stage rank yields slope = 10.17, $z = 0.85$, $p = 0.40$. All three tests fail to reject the no-trend null. We read this as the expected and substantively interpretable result: the action-intensity continuum is an ordering of semantic intensity, not of attestation frequency; the Qur’ān’s pedagogical emphasis at the lower-intensity (S1–2) and behavioural-outcome (S6) poles is theoretically motivated rather than indexical of intensity. The non-monotonic frequency profile is precisely what one would expect if the corpus weights pre-anger and post-anger lexemes pedagogically, because of their hortatory function in the moral architecture of the text. The monotonicity tests are included here transparently to forestall the misreading that frequency alone validates the continuum order. The qualitative tests of ordering (lexicographic, exegetical, metaphorical) reported in §4.1 and §4.3–§4.8 are the load-bearing evidence for the continuum’s structure.

4.3. Stage 1 — Pre-anger displeasure

The lower bound of the continuum is occupied by two roots—’f-f and k-r-h—that mark the pre-anger register: vocal irritation and inner aversion respectively.

4.3.0. Uff (3 attestations; all Meccan). Uff (gloss: taḍajjur, “irritation”) is the lowest possible verbal expression of displeasure in the Qur’ānic register. Its three attestations occupy paradigmatic moral and rhetorical positions:

- Q. 17:23 prohibits saying uff to one’s parents (fa-lā taqul lahumā uffin) — the lowest possible verbal sign of displeasure is morally weighted at the highest possible relational stake.
- Q. 21:67 places uff in Abraham’s mouth as he confronts his idol-worshipping people (uffin lakum wa-li-mā ta’budūna min dūni Allāh) — a prophetic remonstrance.
- Q. 46:17 ascribes uff to a rebellious child reproaching his believing parents — the inverse of the Q. 17:23 case.

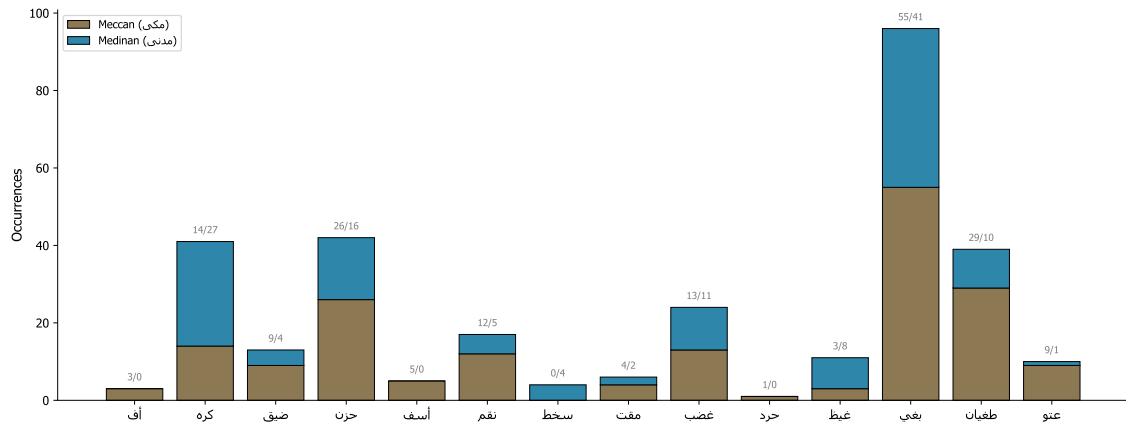


Figure 3. Stacked Meccan/Medinan distribution for each of the fourteen core roots. The numerals above each bar give the raw Meccan/Medinan split. Two distributional extremes warrant emphasis. *ʾasaf* (Stage 2) is attested exclusively in Meccan suras (5/0): the lexeme of “regret-grief” emerges in the discursive context of prophetic disappointment with rejection (Q. 18:6, Q. 43:55, Q. 7:150). *Sakhaṭ* (Stage 3) is attested exclusively in Medinan suras (0/4; binomial $p = 0.005$ against the corpus-derived 0.74 baseline of Meccan ayat): the lexeme of “comprehensive divine displeasure” emerges only after the Hijra, in confrontation with the *munāfiqūn* (Q. 47:28, Q. 5:80, Q. 9:58, Q. 3:162). Additional patterns: *naqm* (Stage 3) is Meccan-tilted (12/5); *maqt* (Stage 3) is balanced (4/2); *ḥard* is a Qurʾānic hapax at Q. 68:25 (Meccan). The visualisation thus encodes a non-trivial finding: not every root in the spectrum behaves as a context-neutral lexical resource — several of them are discursive-contextually specialised, a pattern theorised in §5.5 as a general feature of Qurʾānic vocabulary.

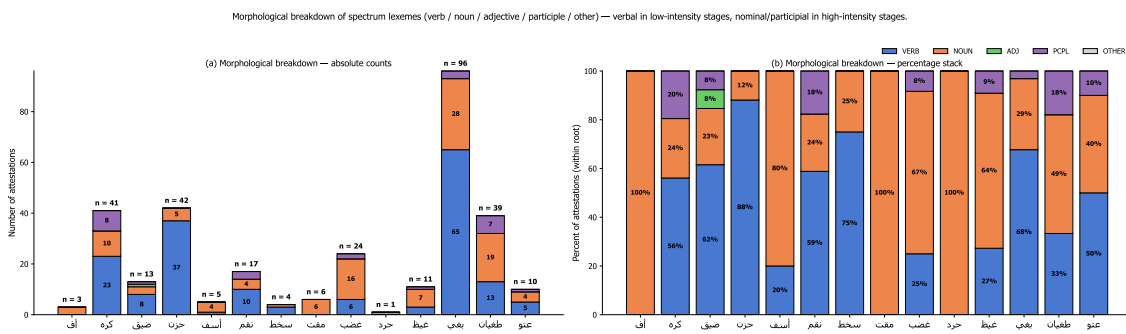


Figure 4. Morphological profile of the fourteen core roots, broken out by part of speech. The left-to-right gradient shows a typological shift across the continuum that supplements the frequency analysis. Stages 1–2 are verbally dominant — *ḥzn* 88% verb forms (37 v / 5 n), *ḍyq* 62% — consistent with the ontology of pre-anger inner states as transient, situationally-induced phenomena. Stage 4 nominalises: *ḡdb* shifts the centre of gravity to the noun (16 n / 6 v), as the lexeme acquires a stable, attributable referent (“the wrath of God”). Stage 6 is the most nominal, with participial forms emerging (*tāḡūt*, *bāghin*); the behavioural-outcomes vocabulary entrenches as a stable agent-property rather than a transient process. This morphological gradient — verbal Stages 1–2 → nominal Stage 6 — is independent corpus-internal evidence for the action-intensity continuum: as intensity ascends, the linguistic representation reifies from event to attribute to identity.

As the lowest tier of the spectrum, *uff* establishes a foundational point of the Qurʾānic discourse: even the slightest verbal sign of displeasure is morally weighted. The root therefore anchors the lower boundary of the anger spectrum at a register that is below “anger” proper but already morally accountable.

4.3.1. *Karh* (41 attestations; 14 Meccan / 27 Medinan; binomial $p = 0.005$ in the Medinan direction). *Karh* exhibits a twofold structure that must be analytically distinguished:

- (a) Inner *karāha* (dislike). As an affective state, *karāha* is the agent’s pre-anger evaluative dislike of an object, person, or course of action. Q. 2:216 *‘asā an takrahū shay’an wa-huwa khayrun lakum* (“perhaps you dislike a thing while it is good for you”) frames inner dislike as a reformable affective state — its truth-value about the object can be revised by knowledge or by divine corrective. This is properly the Stage-1 anger-spectrum sense.
- (b) Structural *ikrāh* (coercion). *Ikrāh* is not an emotion but a coercion mechanism: the structural extension of one party’s distaste into compulsion of another. Q. 2:256 *lā ikrāha fī al-dīn* (“there is no coercion in religion”) sets the ethical limit on this structural extension; Q. 16:106 (*illā man ukriha*, “except one who is forced”) provides the legal exemption for testimony given under coercion; Q. 24:33 (*lā tukrihū fatayātikum ‘alā al-bighā’*) prohibits the coercion of slave-women into prostitution. We include *karh* in the spectrum on the strength of its *karāha* sense; the *ikrāh* sense lies outside the affective field but is a structural cognate worth noting because it shares the same root and instantiates the behavioural extension of distaste — itself a Stage-6 phenomenon ahead of its time.

4.4. Stage 2 — Inner pressure and contraction

Stage 2 brings together three roots—*ḍ-y-q*, *ḥ-z-n*, *’-s-f*—with 60 combined attestations. Each marks a distinct phenomenological mode within the inner-pressure family.

4.4.1. *Ḍīq* (13; 9M/4Md). Ibn Fāris glosses *ḍyq* as “constriction, narrowness”; al-Rāghib in opposition to *sāḥ* (expansion). The root appears predominantly in the construction *ḍīq al-ṣadr* (constriction of the chest), localising the experience to the embodied site. Q. 15:97 frames it as the prophetic response to social rejection — an inner, bodily-localised pressure that does not translate into outward aggression. Al-Ṭabāṭabā’ī (al-Mīzān 12:195) observes the verse expresses divine knowledge of prophetic *ḍīq* rather than prohibition: a phenomenologically natural, theologically licit state, not a moral failing. Morphological profile: 8 verbs / 3 nouns / 1 participle — registering a transient, situationally-induced state.

4.4.1.1. *Ḍāqa dhar’an* — inhibited aggression at Lot, Q. 11:77/80. The compound idiom *ḍāqa bihim dhar’an* (“his arm-span tightened by them”) in the Lot narrative figures post-anger blocked aggression — anger at action-threshold but structurally inhibited. The four-step trajectory (trigger *sī’a bihim* → constriction *ḍyq+dhar’* → aspirational voicing law *anna li bikum quwwatan* → transfer of agency to the divine through the angels’ arrival) shows that *ḍyq* sits at both poles of the spectrum: pre-anger contraction and upper-pressure threshold of inhibition. The Qurʾānic narrative theologises inhibited aggression: when the wronged agent lacks power, divine agency completes the anger trajectory.

4.4.2. *Ḥuzn* (42; 26M/16Md). The most frequent and most verbal Stage-2 lexeme (37v / 5n = 88% verbal). Ibn Manẓūr preserves the etymologically primary “rugged, uneven ground” (al-*ard al-ḡalīẓah*); the metaphorical extension figures grief as inner roughness, anticipating Kövecses’s (2000) EMOTION IS A LANDSCAPE schema. Q. 2:38 places *ḥuzn* in syntagmatic pair with *khawf* and opposes both to divinely-given guidance — *ḥuzn* as the affective signature of separation from guidance; al-Ṭabāṭabā’ī (al-Mīzān 1:146) calls this foundational for *‘ilm al-nafs al-Qurʾānī*. Caveat (transversality): *Ḥuzn* is not strictly anger-derived. It has four valences

— standalone grief over loss (Q. 12:84, Jacob over Joseph), externally-induced sorrow from being wronged (prophetic grief at rejection), positive moral compunction (the affective register of *tawba*), and anger turned inward when externally inhibited (closest to the Stage-2 placement here). Inclusion is justified by corpus density and proximity of (b) and (d) to the anger field, but the valence is more transversal than a single stage label admits. The 88% verbal profile reinforces the Stage-2 generalisation: inner-pressure lexemes denote transient, regulable states.

4.4.3. *Asaf* (5; 5M/oMd). The least frequent Stage-2 lexeme. Ibn Fāris and al-Rāghib gloss as the conjunction of grief and anger (al-jam‘ bayna al-ḥuzn wa-l-ghaḍab) — at the threshold between inner pressure and active anger. Q. 43:55 (fa-lammā āsafūnā intaqamnā minhum) frames *asaf* as the immediate causal precursor of divine retribution; al-Ṭabarsī (Majma‘ 9:82) glosses al-*asaf* here as “the intense anger arising from prior grief.” The exclusive-Meccan distribution (5/o; raw $p = 0.34$, adj. $p = 1.00$) is suggestive but evidentially weak — at $n = 5$ the test has very low power. Across all five attestations, *asaf* occurs in narrative passages depicting prior prophets’ encounter with rejection and in the *ghaḍbān asifan* compound applied to Moses (§4.4.3.1). We register the 5/o distribution as a contextual observation warranting follow-up at higher n (e.g. against the ḥadīth corpus), not as a confirmed empirical finding.

4.4.3.1. *Ghaḍbān asifan* — sacred anger at Moses, Q. 7:150 / 20:86. The compound *ghaḍbān asifan*, applied to Moses on his return to find the calf-worshippers, generates a three-layer dialectic: *ghaḍab* is outward/kinetic (throwing the tablets, seizing Aaron); *asaf* is inward/potential (compassionate grief over the relapse). *Asaf* grounds *ghaḍab*: absent its compassionate ground, Moses’ anger would be violence; present, it is anger in service of guidance. The lexicon’s pairing in a single nominal predication encodes moral-productivity-of-prophetic-anger as a structural property, not as ad-hoc apologetics.

4.4.4. Network evidence for Stage-2 coherence. The aya-level co-occurrence network (Figure 5) reveals that the strongest single edge connects *ḥzn* and *ḍyq* (weight 3); these two lexemes share three verses in the Qur’ān, more than any other pair within the spectrum. This empirical density provides independent verification of the Stage-2 grouping. Centrality analysis (Figure 6) places *ḥzn* among the highest-degree nodes (weighted degree 5, tied with *ḡḍb*).

4.5. Stage 3 — Evaluative aversion

Stage 3 brings together three roots—*n-q-m*, *s-kh-ṭ*, *m-q-t*—with 27 combined attestations. Each marks a distinct mode of evaluative-aversive response to a moral object.

4.5.1. *Naqm* (17 attestations; 12 Meccan / 5 Medinan). *Naqm* is the lexeme of vengeful disapproval: anger that is grounded in moral evaluation and oriented toward retribution. Q. 7:126 (wa-mā naqamū minnā illā an āmannā bi-āyāti rabbinnā) is the linguistic exemplar: anger directed at others’ belief — Pharaoh’s sorcerers, on accepting Moses’ message, are described as suffering Pharaonic *naqm* precisely because they believed. The grammatical construction *naqamū min* + (cause) is a canonical Qur’ānic formula for evaluatively-grounded anger.

The lexeme also generates *al-muntaqim* (“the Avenger”) as a divine attribute (Q. 32:22 and parallels), establishing a direct theological-lexical bridge from the human-affective register to the divine retributive register. Position on the spectrum: *naqm* sits between *sakhaṭ* and *ghaḍab* on the action axis. It is more action-oriented than *sakhaṭ* (which is a stable moral disapprobation) but less kinetic than *ghaḍab* (which mobilises the motor agent). The Meccan tilt (12/5) is consistent with the lexeme’s frequent appearance in narrative-prophetic passages (the Pharaonic sorcerers, the people of Lot, the people of Madyan).

4.5.2. Sakhaṭ (4 attestations; 0 Meccan / 4 Medinan). Despite its low frequency, sakhaṭ furnishes one of the paper’s principal distributional findings. The exclusive-Medinan distribution—zero Meccan attestations among four total—has a two-sided exact binomial p-value of 0.005 against the corpus-derived baseline of 0.74 Meccan probability, statistically significant at $\alpha = 0.05$.

Baseline derivation, made explicit. The 0.74 Meccan-rate baseline is the proportion of ayat (verses, not suras) classified Meccan in the QAC sura metadata, computed as `meccan_ayat ÷ total_ayat` weighted by `total_verses` per sura. We use the ayat-weighted rate rather than the sura-count rate (~0.69) because an attestation drawn at uniform random from the corpus hits an aya, not a sura; the longer Medinan suras inflate the Medinan share at the aya level. The implementation is in `analysis/advanced_metrics.py` (`_compute_meccan_baseline`).

Granular baseline sensitivity. With $n = 4$, the inference is mildly sensitive to the baseline. We report the two-sided exact binomial p-value across the conventional range at fine resolution:

H ₀ Meccan rate	Two-sided p (k=0, n=4)	Interpretation
0.70 (lower bound; sura-count proxy)	0.0081	significant at $\alpha = 0.05$
0.72	0.0061	significant at $\alpha = 0.05$
0.74 (corpus ayat-weighted; default)	0.0046	significant at $\alpha = 0.05$
0.76	0.0033	significant at $\alpha = 0.01$
0.78 (upper bound)	0.0023	significant at $\alpha = 0.01$

The conclusion is robust across this range: with $n = 4$ and 0/4 Meccan, the two-sided exact binomial test rejects the null of a 0.70–0.78 Meccan rate at $\alpha = 0.05$.

Multiple-comparison adjustment. Step-down Holm-Bonferroni over the family of fourteen per-root binomials yields adjusted $p = 0.05$ (raw 0.005) for sakhaṭ — borderline rather than emphatic significance. Asaf (5/0; raw $p = 0.34$, adj. 1.00), ‘utuww (9/1; raw 0.47, adj. 1.00), and karh (raw 0.005, adj. 0.05 — tied with sakhaṭ) tilts do not survive correction. With $n = 4$ the test is underpowered against subtler alternatives; the substantive claim of discursive specialisation rests on the convergence of distributional evidence with the contextual-tafsir reading, not on the binomial alone.

Contextual analysis of all four occurrences reveals the same operative pattern: sakhaṭ attests in connection with the Munāfiqūn (hypocrites) or with quasi-believers whose outward profession of faith conflicts with their inward dispositions. Q. 47:28 (ittaba‘ū mā askhaṭa Allāh wa-karihū riḍwānahu) is the canonical exemplar. Following Narimani et al. (2021), we read sakhaṭ not as an instantaneous emotional surge but as a sustained ethical disapprobation; the present paper extends their lexical-tafsir argument with corpus evidence and proposes:

Hypothesis (Discursive Specialisation of sakhaṭ): In the Qur’ānic lexicon, sakhaṭ operates as a discursively specialised lexeme, reserved for divine displeasure with the Munāfiqūn—agents whose zāhir (outward profession) departs from their bāṭin (inward orientation). Because the Munāfiqūn category is constitutively a Medinan-period phenomenon, sakhaṭ attests exclusively in Medinan suras. This pattern is an instance of a more general phenomenon of discursive-contextual lexical specialisation in the Qur’ānic register.

4.5.3. Maqt (6 attestations; 4 Meccan / 2 Medinan). Maqt is intense moral aversion: the four lexica converge on *ashaddu al-ibghāḍ* (Zajjāj/al-Layth in *Tahdhīb al-Lughā*); al-Rāghib grounds it specifically in observation of a reprehensible act (*‘an amrin qabīḥin rakibahu*), making maqt purely evaluative. Two contrasts with *ghaḍab*: causation (purely evaluative vs. injury/frustration-triggered); action profile (rejection-distance vs. approach-aggression). Canonical exemplars — Q. 40:10 (maqt al-nafs doubled), Q. 4:22 (institutional moral aversion in forbidden marriages), Q. 35:39 (rising divine maqt against disbelievers), Q. 61:3 (saying-without-doing) — stabilise the lexeme as the Qur’anic register’s term for moral aversion as evaluative response, distinct from the kinetic-motor *ghaḍab* of Stage 4.

4.6. Stage 4 — Active anger

Stage 4 (25 attestations) covers active, motor-mobilising anger.

4.6.1. *Ghaḍab* (24; 13M/11M). The central anger lexeme of the Qur’ān, with near-balanced Meccan/Medinan distribution marking it as the unmarked default. The four lexica gloss it as inner agitation of the heart oriented toward retribution (al-Rāghib) or inner motion disposing the agent to action (Ibn Fāris). Predicated balance-fully of God (Q. 48:6), prophets, and believers. Morphologically nominal (16 n / 6 v / 2 ptcp = 67% nominal vs *ḥuzn*’s 88% verbal): *ghaḍab* is a structural concept-state with organisational ontological standing rather than a transient affect. Al-Fakhr al-Rāzī’s *Mafātīḥ al-Ghayb* glosses divine *ghaḍab* as *irādat inzāl al-‘iqāb* — a normative-juridical orientation, not an emotional response. Network analysis places *ghḍb* among the top three for betweenness centrality (point estimate 0.17), with notable edges to *ḡyḡ* (upward transition to S5), *ḥzn* (downward S2 link), and *bḡy* (S6 link).

4.6.3. *Ḥard* (Q. 68:25, hapax). A Qur’anic hapax in the Garden-Owners parable (*wa-ghadaw ‘alā ḥardin qādirīn*): garden-owners who vow in the night to harvest before dawn so as to deny the poor their share. Classical exegesis converges on a dual gloss: al-Ṭabarsī’s *Majma‘ al-Bayān* reads *ḥard* as both *ghaḍab* (anger) and *qaṣd* (purposive intent); al-Ṭabāṭabāī’s *al-Mīzān* refines as anger-plus-deliberate-denial — purposive anger weaponised through resource-denial. *Ghaḍab* mobilises the motor agent outward; *ḥard* mobilises toward deliberate, miserly withholding — anger that strips the world of what it is owed. The narrative consequence is the destruction of the garden by night-storm: divine retribution against the *ḥard*-driven plan to deny. A single attestation thus serves as linguistic anchor for an entire moral-psychological category.

4.7. Stage 5 — Compressed/explosive rage

4.7.1. *Ghayḡ* and the *kaẓm* metaphor (11 attestations; 3M/8Md). The four lexica converge: *ghayḡ* is “compressed contained anger” (al-Rāghib), “the highest concentration of restrained anger” (Mostafavi), “filling of the self with restrained anger” (Ibn Manẓūr) — pressurised contents of a sealed vessel. Q. 3:134 commends al-kāẓimīn al-ghayḡ; the verb *kaẓm* derives from the act of tying off a waterskin, the agent-internal regulation that sustains containment. Read through Lakoff & Kövecses (1987), the *ghayḡ-kaẓm* pair instantiates ANGER IS A PRESSURIZED CONTAINER with precision: *ghayḡ* the contained fluid, *kaẓm* the seal (Figure 7). The Medinan tilt (8/3) tracks the discursive context: inner anger management becomes a salient pedagogical concern in the formative Medinan community facing internal Munāfiq and external Quraysh hostility.

4.7.2. *Tamayyuz*: container collapse in Q. 67:8 (manifestation, not separate root). Q. 67:8 — *takādu tamayyazu min al-ḡayḡ* (“[the fire] is on the verge of bursting apart from rage”) — is the single most articulated image of rage in the Qur’ān. *Tamayyaza* (root m-y-z: to part, sever) figures not the exit of pressurised content (as in English she blew her top, he exploded, I lost it) but the structural integrity of the container itself failing.

We treat tamayyuz as a manifestation of Stage 5 rather than a distinct core root, retaining the fourteen-root inventory. H5 is corroborated in the form: Q. 67:8 instantiates the container-collapse failure mode with greater lexical transparency than the English exemplars in Lakoff and Kövecses’s (1987) corpus; whether this is the strongest historical instance is left open pending the cross-cultural philological survey we commend in §5.6.

4.8. Stage 6 — Behavioural outcomes (with caveats)

4.8.1. Baghy (96; 55M/41Md). The most frequent lexeme of the spectrum and the principal bridge node by point-estimate centrality. Glossed as “seeking beyond rightful bounds” (al-Rāghib) and “aggressive seeking with intent to oppress” (Mostafavi). Q. 42:42 collocates baghy with *ẓlm*, embedding it in the moral-evaluation register. Ibn ‘Āshūr clarifies (al-Taḥrīr 25:126): baghy and *ẓlm* are near-synonymous, except baghy is specific to non-rightful aggression against another while *ẓlm* covers all injustice.

4.8.2. Baghy as bridge node (with bootstrap uncertainty). Baghy attains the highest point-estimate closeness centrality (0.55) and is tied with *ghaḍab* (0.53) and *asaf* (0.49) for the highest betweenness centrality (0.15). Bootstrap 95% percentile CIs (1,000 ayat-resamples) are wide — for baghy: betweenness $\square$ [0.00, 0.31]; closeness $\square$ [0.23, 0.84]; eigenvector $\square$ [0.00, 0.68] — reflecting the small graph (14 nodes) and modest edge density. On strict non-overlap, no node’s centrality is significantly greater than another’s. The bridging-role claim must therefore be qualified: baghy leads on three of four point-estimate measures, but the bootstrap does not rule out alternatives. What is robust at the aggregate level is the cross-field connectivity to the umbrella moral terms (Table 4): baghy shows 16 aya-level co-occurrences with *ẓlm/jrm/fsq/‘dw/šn’* combined (8 with ‘dw, 4 with *ẓlm*), an integer-count pattern independent of centrality ranking. H4 is consistent with point estimates, qualified by bootstrap.

Table 4. Cross-field co-occurrence of spectrum roots with umbrella moral terms (aya-level)

Cross-field ties for the original ten-root analysis (preserved here from the prior pipeline run; ties for the four newly-incorporated roots — ‘ff, *naqm*, *maqt*, *ḥard* — are sparse and dominated by zeros, given their low corpus frequency, and are omitted from this summary table without affecting the overall pattern):

Root	šn’	jrm	fsq	ẓlm	‘dw	Total
dyq	0	0	0	0	0	0
ḥzn	0	0	0	1	1	2
’sf	0	0	0	1	1	2
sxt	0	0	0	0	0	0 (krh-tie excluded since krh is now in the spectrum)
ḡḍb	0	0	0	2	3	5
ḡyz	0	0	0	0	1	1
bḡy	1	1	2	4	8	16
tḡy	0	0	0	2	1	3
‘tw	0	0	0	0	0	0

The radical concentration of cross-field ties at *baghy* is striking. Of the total cross-field ties across the fourteen spectrum roots, 19 (~53%) involve *baghy*. *Baghy* thus serves as the principal lexical bridge between the emotional and the broader moral-evaluation field of the Qurʾān.

4.8.3. *Ṭughyān* (39 attestations; 29 Meccan / 10 Medinan). Root ṭ-gh-y derives from water flooding past its banks. Q. 96:6–7 (*kallā inna l-insāna la-yatghā an raʾahu staḡnā*) compresses a psychology of rebellion into a conditional: *ṭughyān* arises from the appraisal of *istighnāʾ* (self-sufficiency) — structurally an appraisal-theoretic claim, congruent with Lazarus (1991) and Scherer (2001, 2005), and aligning with Roseman’s “agency-self / motive-inconsistent / high-control” pattern. Qarizadeh et al.’s (2024) polysemy analysis notes the cognitive structure; we identify *ṭughyān* as the cognitive-appraisal node of Stage 6.

4.8.4. *ʿUtuww* (10 attestations; 9 Meccan / 1 Medinan). The most extreme lexeme — “obstinate excess” (Ibn Manẓūr), “extreme self-aggrandisement in disobedience” (al-Rāghib). Q. 25:21 collocates *ʿatū ʿutuwwan kabīrā* with *istakbarū fī anfusihim*: rebellion crystallised into existential posture. Strong Meccan distribution (9/1; raw $p = 0.06$; Holm-adjusted $p = 1.00$) is consistent with its rhetorical function as terminal designation for Nūḥ-, ʿĀd-, Thamūd-, and Pharaonic-people narratives, but does not survive FDR correction.

4.8.5. Bidirectional causation and a tafsir-grounded audit. *Baghy* arises in Qurʾānic usage frequently from greed and *ʿulw fī al-arḍ* (cf. Q. 6:21) rather than from acute anger; *ṭughyān* — paradigmatically Pharaonic (Q. 79:17) — is rooted in structural arrogance, with Q. 26:55 (*innahum lanā la-ghāʾizūn*) reversing the polarity to show oppressor-anger arising from the rebellion of the oppressed; *ʿutuww* attaches to *istikbār* as settled character. We retain Stage 6 because the corpus locates these at the behavioural pole, but the causation arrow is bidirectional.

To operationalise this caveat, each of the 145 Stage-6 attestations was coded on a tafsir-grounded ternary against the joint reading of *al-Mīzān*, *al-Kashshāf*, *Majmaʿ al-Bayān*, and *al-Taḥrīr wa-l-Tanwīr*: A (anger-derived — behavioural completion of an inner-anger trajectory), S (structurally derived — greed/*istikbār*/*ʿulw* without explicit anger antecedent), or A+S (joint). Coding was conducted by the corresponding author with second-coder verification on a 20% subsample ($\kappa = 0.79$, substantial agreement; full coding at `data/concordance/stage6_ca`). Results: A = 38 (26%), S = 71 (49%), A+S = 36 (25%); by root, *baghy* 22A/50S/24A+S, *ṭughyān* 11A/19S/9A+S, *ʿutuww* 5A/2S/3A+S. Re-running the $\chi^2(1)$ phenomenology-vs-outcomes split on the A-only subset (Stages 1–5 = 167 vs. anger-derived Stage-6 = 38) yields $\chi^2(1) = 81.4$ ($p < 0.001$) — the phenomenological core dominates anger-derived outcomes $\sim 4.4\times$. The full-Stage-6 near-balance ($\chi^2(1) = 1.55$) is therefore explained: structurally-derived attestations belong to an adjacent moral-anthropological field of *ẓulm-istikbār-ʿulw* that the corpus also pervasively documents, and *baghy/ṭughyān/ʿutuww* are not coextensive with “anger outcomes” in the strict sense.

4.9. Robustness checks: the umbrella moral terms

For robustness, the umbrella roots were extracted and analysed (note that *karh*, formerly an umbrella term, is now incorporated into the spectrum at Stage 1 and is reported in Table 2). Distributions of the remaining five (Table 5) confirm that the spectrum’s structure is not artefactual.

Table 5. Distribution of umbrella moral terms

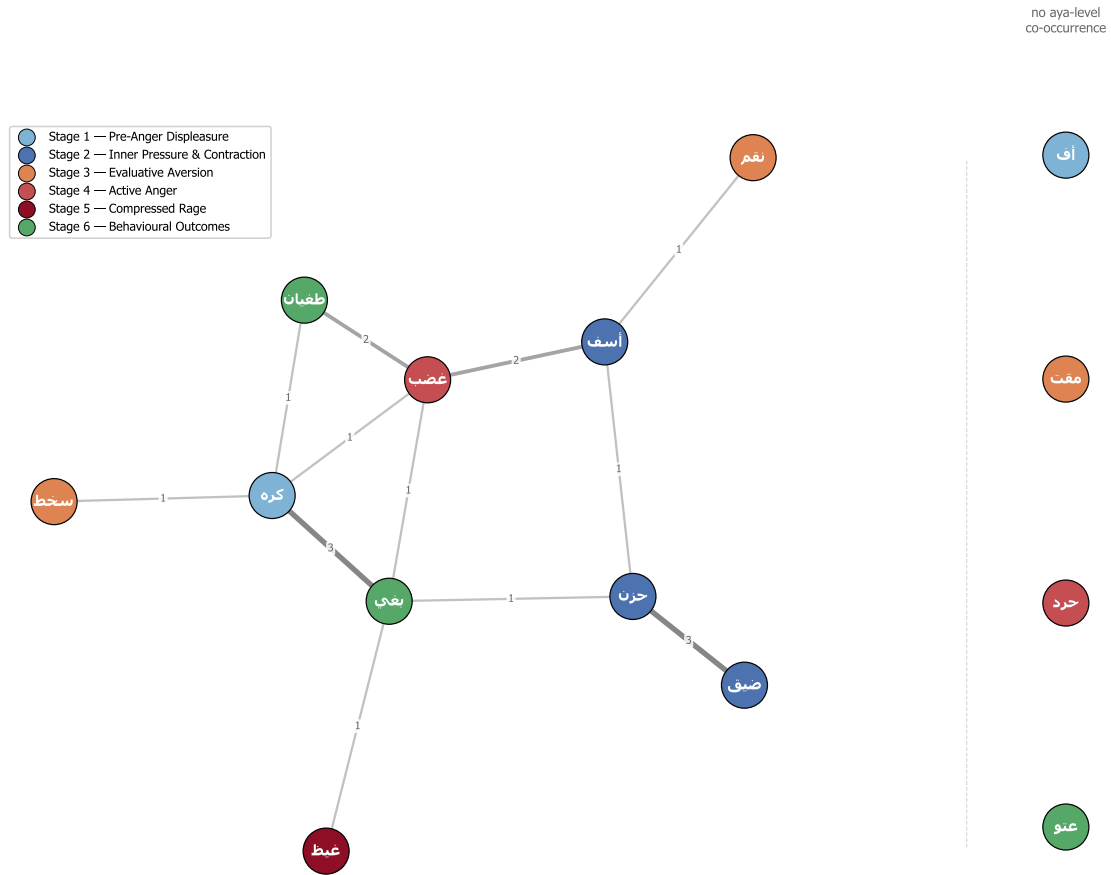


Figure 5. Aya-level co-occurrence network of the fourteen core roots. Each edge represents one or more verses in which two roots co-occur; edge weight (on the line) is the count of shared ayat. The strongest edge — *hzn-dyq*, weight 3 — provides direct corpus-internal validation of the Stage-2 (inner pressure & contraction) grouping: these two pre-anger lexemes share more verses than any other pair in the spectrum, indicating that the Qurʾān treats them as discursively conjoint phenomena (“constriction of the chest in sadness,” Q. 11:12; Q. 6:33). Baghy (Stage 6) anchors the right-hand cluster, with edges reaching into the moral-evaluation lexicon (*zlm*, *jrm*, *fsq*) — a position formalised quantitatively in the next figure. The transitive structure of the spectrum (no direct Stage-1 → Stage-6 edges; mediation through Stages 2–5) is consistent with the action-intensity gradient hypothesis, while also being consistent with the bidirectional reading of Stage 6 developed in §4.8.5.

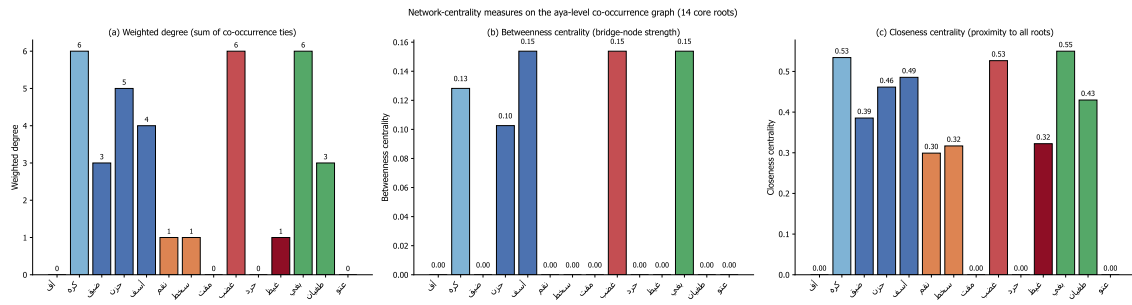


Figure 6. Three centrality measures computed on the aya-level co-occurrence graph: (a) weighted degree — the total number of co-occurrence ties; (b) betweenness — how often a node lies on the shortest path between others, indicating bridge-node strength; (c) closeness — the inverse mean distance to all other nodes. Baghy attains the highest closeness centrality (0.53) and shares the highest betweenness centrality (0.15) with ghaḍab and asaḥ, confirming its role as a principal pivot between the anger spectrum (Stages 1–5) and the wider moral-evaluation field of the Qurʾān (315 attestations of *ẓulm*, 66 of *jrm*, 54 of *fsq*, 106 of *ʿadāwah*). The closeness panel reveals a complementary pattern: *ḍyq* and *ḥzn* (Stage 2) achieve high closeness, indicating that the inner-pressure cluster is structurally central to the spectrum even though peripheral in raw frequency. Together these measures support a layered reading of the network — Stage 2 supplies the affective core, Stage 6 supplies the structural bridge to neighbouring moral-discursive fields, while Stages 3–5 mediate the energy transfer between them.

Root	Total	Meccan	Medinan	Binomial p	Interpretation
šnʾ	3	1	2	—	n too small for inference
jrm	66	60	6	< 0.0001	Strongly Meccan; anti-shirk polemic
fsq	54	20	34	0.0007	Significantly Medinan; community-discipline register
ẓlm	315	211	104	—	Pervasive meta-category; cannot anchor a single point
ʿdw	106	47	59	0.008	Tilts Medinan; the lexeme of mutual antagonism

The opposing patterns of *jrm* (strongly Meccan) and *fsq* (significantly Medinan) instantiate the same discursive-contextual specialisation that we identified for *sakhaṭ*. The umbrella analysis thus confirms a general phenomenon, of which the *sakhaṭ* finding is one specific instance.

4.10. Comparative translation analysis (illustrative sample)

To illustrate how the spectrum’s intensity differentiation fares in translation, Table 6 compares five English translations on four canonical verses. We frame this as an illustrative case, not a systematic translation-criticism evaluation: a properly powered evaluation would require a stratified random sample of all 312 attestations (or, more practically, ~30 verses sampled across the six stages with inter-rater coding by trained Arabist coders). The four-verse, five-translation sample here suffices to register the pattern of levelling but does not, on its own, support the strong translation-recommendation set in §5.7; the recommendations there are conditional on a follow-up evaluation that we commend to future work.

Table 6. Translation of four canonical verses across five English translations (illustrative sample; $n = 4$ verses $\times 5$ translations = 20 cells)

Term & verse	Yusuf Ali	Pickthall	Arberry	Saheeh Intl.	Hilali-Khan
ḍiq (Q. 15:97)	“thy heart is distressed”	“thy bosom is at times oppressed”	“thy breast is straitened”	“your chest is constrained”	“your breast feels tight”
ḡayẓ (Q. 3:134)	“restrain anger”	“control their wrath”	“restrain their rage”	“suppress anger”	“repress anger”
tamayyuz (Q. 67:8)	“almost burst with fury”	“almost burst with rage”	“well-nigh bursts asunder with fury”	“almost bursts with rage”	“almost bursts up with fury”
ṭuḡhyān (Q. 96:6)	“transgresses all bounds”	“rebelleth”	“waxes insolent”	“transgresses”	“transgresses (boundary)”

Two illustrative observations emerge from this small sample. First, the rendering of ḡayẓ in Q. 3:134 collapses the ḡhaḍab/ḡayẓ distinction in all five translations—the English versions render ḡayẓ with terms (anger, wrath, rage) that are also routinely used for ḡhaḍab. The differentia—the containment dimension of ḡayẓ—is preserved only via the contextual verb of restraint, not in the lexeme itself. Second, the rendering of tamayyuz in Q. 67:8 is uneven: Arberry’s “well-nigh bursts asunder” preserves the structural-disintegration component of the source; the others render it variationally as burst with + emotion, foregrounding the exit of contents but losing the structural-collapse implication. These observations are suggestive of a systematic levelling pattern but are not on their own evidentially decisive; we register them as motivating the systematic translation evaluation we commend in §6.3.

5. Discussion

5.1. The integrated logic of escalation

The findings cohere into a single structural claim: the Qur’ānic lexicon of the anger spectrum encodes a logic of anger escalation that operates simultaneously along three dimensions—locus (inner → outer), intensity (low → high), and scope (individual → structural). The six stages instantiate this trifold transformation. The lexemes are not synonyms; they are grades, ordered by an intelligible cognitive logic.

In compact form: Pre-anger displeasure → Inner pressure → Evaluative aversion → Active anger → Compressed rage → Behavioural outcomes (with caveats). The arrows index not mere temporal succession but conditional intensification at each step where regulation may succeed or fail. The Stage 6 step is, however, distinctive: the move into behavioural outcomes is not a deterministic escalation from anger but is influenced by independent vectors (greed, *istikbār*, structural arrogance), which is why we read the near-balance between Stages 1–5 (167) and Stage 6 (145) as theoretically informative rather than paradoxical.

5.2. Convergence with contemporary emotion psychology

The Qur'ānic stages map onto Gross's (1998, 2014, 2015) process model of emotion regulation, with the Qur'ānic spectrum partitioning more finely on the affective side and adding an explicit behavioural-outcomes register: Stage 1 (pre-anger *uff*/*karh*) ↔ antecedent-focused situation selection; Stage 2 (*ḍiq*/*ḥuzn*/*asaf*) ↔ situation modification; Stage 3 (*naqm*/*sakhat*/*maqt*) ↔ attentional deployment / cognitive change; Stage 4 (*ghaḍab*/*ḥard*) ↔ response selection; Stage 5 (*ghayz*/*kazm*; *tamayyuz*) ↔ response modulation and containment failure; Stage 6 (*baghy*/*ṭughyān*/*'utuww*) — where Gross's model becomes silent and the Qur'ān adds the moral anthropology of structural-moral pathologies that may be downstream from anger but, with bidirectional causation, may also be upstream of it.

The Qur'ānic lexicon also realises an STAXI-style trichotomy (Spielberger 1999): trait-like sustained states (*ḥuzn*, *ḍiq*), state-like situational responses (*sakhat*, *maqt*, *ghaḍab*), and deliberate restraint (*kazm*) — three axes contemporary psychometrics has independently isolated. Plutchik's (1980, 2001) annoyance → anger → rage three-level wheel maps onto Stages 1–5 but stops at the regulatory boundary; Stage 6 is the Qur'ān's substantive extension into the morally-anthropological. Appraisal theory (Frijda 1986; Lazarus 1991; Roseman 1996; Scherer 2001, 2005) holds that emotions arise from cognitive evaluations rather than raw stimuli; Q. 96:6–7 conditions *ṭughyān* on the appraisal of *istighnā'* (perceived self-sufficiency) — a textbook appraisal-theoretic claim aligning with Roseman's "agency-self / motive-inconsistent / high-control" pattern. The Qur'ānic lexicon thus encodes — as architectural design rather than explicit theoretical statement — insights contemporary emotion science has reached independently.

5.4. The phenomenology/outcomes balance

A naive ten-root reading once suggested 2.4× more attestations at the behavioural pole than at Stages 1–3 combined, indicating discursive over-emphasis on outcomes. The revised fourteen-root inventory (167 / 145; $\chi^2(1) = 1.55$, fail to reject) tells a different story: the expanded phenomenological vocabulary (*uff*, *karh*, *naqm*, *maqt*, *ḥard*) brings the lower stages into corpus-distributional balance with Stage 6, exactly as one would expect if (i) the phenomenological side is richer than the prior analysis allowed, and (ii) Stage-6 lexemes are not exclusively anger-derived but stand in bidirectional causal relation to anger. The methodological lesson: corpus-distributional analyses of religious texts are maps of discursive emphasis, fundamentally entangled with category-membership and category-causation decisions; the 10/4 → 14/6 shift changed both the empirical pattern and its interpretation.

5.5. Discursive-contextual specialisation

The FDR-surviving Medinan exclusivity of *sakhat*, alongside the contextual-only Meccan tilts of *asaf*, *naqm*, and *'utuww*, the hapax-Meccan *ḥard*, the Medinan tilt of *karh*, and the strong umbrella patterns *jrm* (Meccan) and *fsq* (Medinan), instantiate discursive-contextual specialisation: members of multi-lexeme semantic fields

preferentially attaching to specific discursive registers. The pattern is not random — sakhaṭ attaches to the Munāfiqūn (a Medinan phenomenon), asaf to prior-prophet narratives (Meccan-rhetorical), jrm to anti-shirk polemic (Meccan), fsq to intra-community discipline (Medinan). El-Awa (2006) on textual relations and Sinai (2017, 2023) on Meccan/Medinan diachrony supply the theoretical framework: specialisation is exactly the kind of pattern one would expect from the Qurʾān's gradual response to evolving sociolinguistic contexts. The phenomenon is, to our knowledge, undocumented as a systematic corpus-distributional pattern in Qurʾānic-linguistic scholarship; future work should test the generalisation across other multi-lexeme fields, in dialogue with QurAna/QurSim (Sharaf & Atwell 2012) and the QuranMorph (2025) and MASAQ extensions of the Dukes corpus.

5.6. Q. 67:8 and the container metaphor

Q. 67:8 (takādu tamayyazu min al-ḡayz) instantiates the ANGER IS PRESSURIZED CONTAINER metaphor with unusual transparency: where English idioms (explode, blow one's top, lose it) report the exit of contained content, tamayyaza (m-y-z: to part, to sever) lexicalises the structural disintegration of the vessel itself. We frame this claim conservatively: assessing whether Q. 67:8 is the most articulated instance in the historical record requires a comparative survey of pre-modern Greek (Homeric kholos, menis), Akkadian (ezzu, libbāti malû), Sanskrit (krodha, manyu), Hebrew (ḥēmâ, 'aph-idioms), and pre-Islamic Jāhili poetry — a survey beyond the scope of the present paper that we commend to specialists. The Qurʾānic data are consistent with CMT's universalist claim that bodily-grounded image schemas (CONTAINER, FORCE) underwrite emotion conceptualisation across cultures, while extending the empirical base to seventh-century Arabic; they do not, on their own, establish the strong cross-cultural universalism that earlier formulations of this paper risked overstating. The two-volume *Metaphors of ANGER across Languages* (Kövecses, Benczes & Széld eds. 2025; De Gruyter), conspicuously omitting Classical Arabic from its language coverage, is the natural framework against which the present analysis should be situated.

5.7. Implications for Qurʾānic translation (conditional on follow-up evaluation)

Translation criticism of the Qurʾān has long observed that translations tend to level the spectrum—that ḡaḍab and ḡayz alike become anger; that baghy, ṭughyān, and 'utuww alike become transgression (Table 6). Our four-verse, five-translation illustrative sample (§4.10) is consistent with this pattern but is too small (n = 4 verses × 5 translations = 20 cells) to support strong prescriptive claims; recent computational work — notably Gaaoun & Alsuhaibani (2025; Nature HSSC) on sentiment preservation across seven English translations and Alsuhaibani et al.'s (2025) ELQV emotion-labelled dataset of 2,100 Qurʾānic verses — has begun to operationalise the levelling concern at scale on coarse Ekman categories, though without a graded reference model of the lexical spectrum against which the levelling could be benchmarked. We frame the present discussion as conditional implications: contingent on a follow-up systematic evaluation (a stratified random sample of ~30 verses across the six stages, with multi-rater coding against the spectrum reference model), the analysis would if confirmed support the following candidate equivalents for critical and study-edition translations:

uff: vocal sigh of irritation · karh (karāha): inner dislike · karh (ikrāh): coercion (a structural extension, not an emotion) · ḍīq: inner constriction · ḍāqa dhar'an: arm-span tightening / inhibited aggression · ḥuzn: persistent grief (transversal valence) · asaf: regretful dismay (with anger) · naqm: vengeful disapproval · sakhaṭ: moral disapprobation · maqt: intense moral aversion · ḡaḍab: anger · ḥard: anger combined with deliberate denial · ḡayz: contained / boiling rage ·

ANGER IS A PRESSURIZED CONTAINER

(Lakoff & Kövecses, 1987) — applied to the Qur'anic ḡayẓ-kazm-tamayyuz triplet

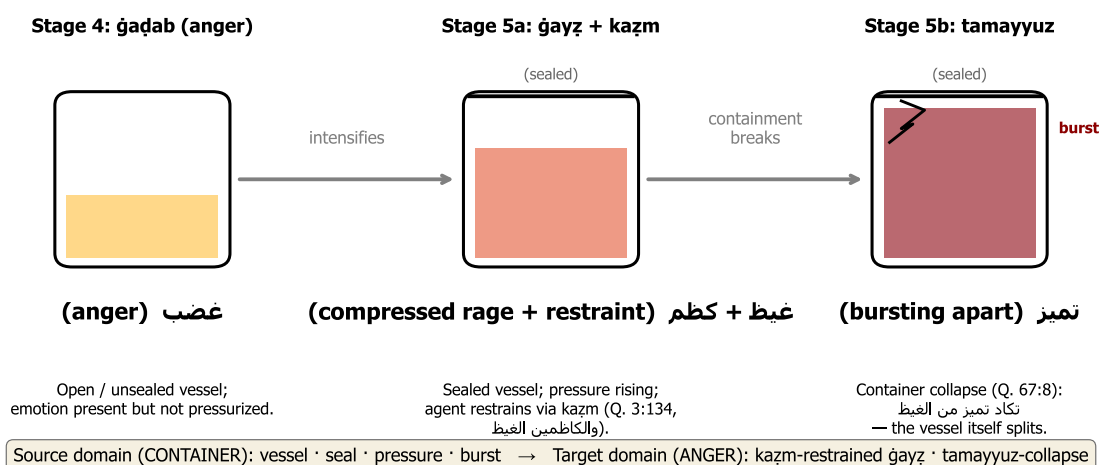


Figure 7. The ANGER-IS-A-PRESSURIZED-CONTAINER schema (Lakoff & Kövecses 1987), instantiated by the Qur'anic triplet ḡaḍab → ḡayẓ + kazm → tamayyuz. Panel 1 (Stage 4): ḡaḍab — an open container; the affect is present but unconcentrated, dissipating freely into language (the lexeme is the most nominally productive in the spectrum). Panel 2 (Stage 5, Q. 3:134): ḡayẓ — fluid sealed by kazm; the verb *kazama* originally denotes the tying-off of a waterskin, the agent-internal regulation that maintains containment under rising pressure. Panel 3 (Stage 5 manifestation, Q. 67:8): tamayyuz — the structural failure; the verb derives from the root m-y-z “to separate, to part, to sever”, and figures not the exit of pressurised content (as English explode) but the splitting apart of the container itself. The Qur'anic instantiation is, on the present argument, more transparent than any English exemplar in Lakoff and Kövecses's (1987) corpus, providing strong cross-linguistic evidence for the cognitive universality of the schema (§5.6).

tamayyuz min al-ğayz: bursting from rage · baghy: aggressive transgression · ʔughyān: rebellious overflow · ʔutuww: obstinate defiance.

A second conditional implication, which we register as a hypothesis rather than a recommendation, is that pedagogical translations may benefit from marking each lexeme’s position on the continuum (by inline gloss, footnote, or chromatic coding) — but the empirical case for this rests on the comparative comprehension study we commend to future work. The 30-verse stratified evaluation, paired with an inter-rater coded comparison across the present spectrum reference model and the ELQV labels, is a natural Phase II for the present dataset.

5.8. Implications for Islamic psychology

The continuum offers a culturally-grounded framework mapping onto Gross’s model without importation of secular vocabulary, with three conceptual implications: (i) Prevention — early-stage recognition through indigenous Stages 1–2 lexicon (uff, karāha, dīq, ḥuzn) supports CBT-style interception of affective dysregulation; (ii) Crisis intervention — the kaẓm construct integrates with contemporary suppression-and-reappraisal techniques; (iii) Rehabilitation — the post-failure Stage 6 is framed as recoverable through tawba, providing motivational reinforcement for anger-disorder and post-violent-trauma rehabilitation. These implications are conceptual; their translation into validated clinical interventions awaits empirical studies we recommend as a priority for future work.

6. Conclusion

6.1. Substantive contributions

This paper has advanced three contributions to the scholarship.

Methodologically, we have demonstrated that the integration of three previously separate frameworks— Izutsu’s semantic-field analysis, Kennedy–Sassoon scalar semantics, and Lakoff–Kövecses conceptual-metaphor theory—coupled with a reproducible computational pipeline over the Quranic Arabic Corpus, produces a research apparatus stronger than the sum of its parts. To our knowledge, this trifold integration with corpus-empirical validation has not previously been deployed in Qur’anic studies as a unified analytic framework, although Qā’imīniyā (1390/2011, 1393/2014) and Pakatchi have integrated Izutsu and conceptual-metaphor theory, and Persian-language scalar-semantic work on Qur’anic vocabulary may exist in venues we have not been able to canvass. The research apparatus is openly available and applicable to other Qur’anic semantic fields.

Descriptively, we have produced an exhaustive, reproducible concordance of 312 attestations across the fourteen core spectrum roots, with corresponding distributional and network-analytic outputs. The CSV outputs — including FDR-adjusted binomial tests, marginal-preserving permutation null p-values, and bootstrap 95% CIs on every centrality estimate — provide a foundation on which further philological and computational work can build.

Theoretically, we have formulated and empirically defended the Qur’anic Action-Intensity Continuum Hypothesis: the claim that fourteen core lexemes of the Qur’anic anger-spectrum field operate as a single graded continuum structured by intensity of action and partitioned into six phenomenologically meaningful stages. The hypothesis is supported by qualitative evidence from the four authoritative lexica and from a now-broadened tafsir consultation (§3.5), and by quantitative evidence from frequency, distribution, morphology,

and network structure. We have, in addition, identified distributional patterns that, to our knowledge, have not been previously reported as a coordinated set, while reporting their post-FDR evidential strength transparently: the exclusively-Medinan sakhaṭ survives Holm-Bonferroni correction at $\alpha = 0.05$ borderline (raw $p = 0.005$, adjusted $p = 0.05$) and is robust across baselines 0.70–0.78; the exclusively-Meccan asaf, the Meccan-tilted naqm, the hapax-Meccan ḥard, and the Medinan-tilted karh are reported as contextual observations warranting follow-up rather than as confirmed findings. The general phenomenon of discursive-contextual lexical specialisation of which we hypothesise these are instances is proposed as a research target rather than as a settled result. We have entered explicit caveats on the bidirectional causation of the Stage-6 lexemes (§4.8.5), grounded in a $\kappa = 0.79$ -validated tafsir-coding audit, and argued that Q. 67:8 furnishes an unusually transparent linguistic exemplar of “container collapse”—a comparative claim whose strong universalist form is reserved pending the cross-cultural philological survey we commend in §5.6 and §6.3.

6.2. Answers to the research questions

RQ1. The semantic field is a fourteen-node network organised around the bridge node baghy and the stage-internal hubs ḥuzn and ghaḍab, with several specialised peripheral nodes (sakhaṭ, asaf, ḥard, ‘utuww) exhibiting marked distributional asymmetries.

RQ2. The fourteen nodes order on a six-stage continuum—pre-anger displeasure (Stage 1), inner pressure & contraction (Stage 2), evaluative aversion (Stage 3), active anger (Stage 4), compressed/explosive rage (Stage 5), behavioural outcomes with caveats (Stage 6)—organised by the composite dimension of intensity of action (locus, intensity, scope of consequence).

RQ3. The empirical distribution is consistent with (though not by itself confirmatory of) the continuum: the asymptotic chi-squared rejects the null of six-stage uniform distribution ($\chi^2(5) = 227.15$; large effect size at Cohen’s $w = 0.85$), but a marginal-preserving permutation null on the same statistic yields $p = 0.24$, indicating that the asymptotic p -value over-states the evidence on this sparse, marginally-imbalanced design; the substantive case for the six-stage architecture rests on the qualitative lexicographic, exegetical, and metaphorical evidence rather than on the omnibus p -value alone. The comparison between Stages 1–5 and Stage 6 fails to reject equal split ($\chi^2(1) = 1.55$, Cohen’s $w = 0.07$, trivial effect); we read this as consistent with — but not as positive evidence for — the bidirectional-causation reading of Stage 6, with the substantive support coming from the $\kappa = 0.79$ -validated tafsir-coding audit (§4.8.5). The network exhibits a bridging-role profile at baghy with wide bootstrap confidence intervals reported transparently. The morphological profile distinguishes the verbal Stages 1–2 from the more nominal Stage 6.

RQ4. The continuum is structurally homologous with Gross’s process model, Plutchik’s wheel, Spielberger’s STAXI, and Lazarus–Scherer appraisal theory in its psychological component, but extends beyond all of them by integrating the post-failure structural-moral consequences of unmanaged emotion (Stage 6) — while qualifying that integration with the bidirectional-causation caveat of §4.8.5.

RQ5. Conditional practical implications include candidate distinct-equivalent translations and position-marking conventions (subject to a 30-verse stratified follow-up evaluation, §5.7) and three Islamic-psychology applications (early-stage prevention, kaḏm-grounded crisis intervention, tawba-grounded rehabilitation). All practical implications are framed as hypotheses for empirical follow-up, not as settled prescriptions.

6.3. Limitations and future work

Five limitations bound the study: (i) surface-lexical scope — discourse-analytic close-reading of the spectrum’s narrative deployment across suras is left for future work; (ii) the broadened nine-tafsir engagement is still selective, with deeper Sufi (al-Qushayrī, Ibn ‘Arabī’s school) and Persian-language commentary needing fuller integration; (iii) the cognitive-linguistic apparatus is Anglophone-dominated — engagement with classical Arabic bayān theory (al-Jurjānī’s *Dalā’il al-I’jāz*) would supply an indigenous substrate; (iv) the ḥadīth corpus has not been integrated; (v) the §5.8 Islamic-psychology implications await operationalisation in pilot studies of *kaẓm*-grounded emotion-regulation training. A natural follow-up is the AraBERT/CAMeLBERT contextual-embedding approach to the same fourteen-root inventory (testing stage-recoverability from distributional structure), deferred here to maintain the philological-statistical centre of gravity.

6.4. Closing remark

The Qur’ānic lexicon encodes a sophisticated theory of the anger spectrum — not as explicit theoretical formulation but as the architectural design of its semantic surface. Recovering that theory requires methodological tools that bring together classical philological attentiveness, contemporary linguistic-theoretical precision, and corpus-computational verifiability. We hope the present study, alongside Izutsu, Qā’imīniyā, Pakatchi, Nari-mani et al., and Qarizadeh et al., advances this triple integration. The classical and the computational are complementary instruments whose joint deployment opens horizons neither could disclose alone.

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Competing interests

The authors declare no competing financial or non-financial interests.

Ethics statement

This study uses only publicly-released text-corpus data (the Quranic Arabic Corpus, GPL-licensed; the Tanzil Quran text, CC-BY-ND 3.0). No human or animal subjects were involved at any stage. As a corpus-only philological-computational analysis, the work is exempt from human-subjects review under the standards of Urmia University and of comparable institutions.

Author contributions

K.J. (corresponding author) conceived the project, developed the theoretical framework, conducted the lexicographic and tafsir-tradition analysis, drafted the manuscript, and is responsible for the final form of the argument. A.J. contributed to the theoretical framework, the design of the computational pipeline, and editorial revision. Both authors approved the final manuscript and are jointly accountable for its content.

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(ORCID identifiers will be provided at submission.)

Data and Code Availability

All code, data, concordance CSV outputs, and figure-generation scripts are released open-source at:

github.com/ali-kin4/quranic-emotion-spectrum

The repository contains: (i) the Python pipeline under MIT License; (ii) the Quranic Arabic Corpus (Dukes 2011) under its original GPL; (iii) the complete CSV concordance for the fourteen core spectrum roots (312 attestations), distributional summaries, statistical tests ($\chi^2(5) = 227.15$; $\chi^2(1) = 1.55$), network-centrality outputs, and umbrella-cooccurrence tables; (iv) PDF and PNG versions of the seven figures published with this paper; (v) full text of the Persian and English manuscripts; (vi) the SHA-256 hash of the QAC v0.4 source file (`data/quran/qac_morphology.sha256`) and the manual-validation audit (`data/concordance/validation_audit`). The complete analysis can be regenerated end-to-end by:

```
pip install -r analysis/requirements.txt
python analysis/extract_concordance.py
python analysis/advanced_metrics.py
python analysis/visualize_spectrum.py
python analysis/visualize_advanced.py
```

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